input trace. Correlation support in AWS X-Ray SDK for Java (http://mng.bz/KeaE) is an example.

11.1.3 Correlating metrics

The starting point of correlation paths in this case are time series of metrics—numerical aggregated values. From here we can transition to specific correlations.

TO TRACES

Metrics–traces correlation means you have a metric as input (e.g., graphed on a dashboard or from an alert), and the expected output consists of representative spans from services contributing to said time series. We will have a concrete look at such a correlation case in the next subsection, where we will use Prometheus, Jaeger, and Grafana to implement exemplar-based correlation between a metric and spans.

TO METRICS

Metrics–metrics correlation means related metrics (by type or service) (e.g., client-side and server-side errors).

TO LOGS

Metrics–logs correlation means the input is a metric (e.g., from a dashboard) and the output consists of log lines from services contributing to the aggregate value.

11.1.4 Correlating logs

The starting points of correlation paths in this case are logs from one or more services. From here we can transition to specific correlations.

TO TRACES

Logs–traces correlation means we have log lines of a service as an input and seek corresponding spans emitted from the service that wrote the log lines.

TO METRICS

Logs–metrics correlation uses log lines of a service as an input and outputs relevant metrics from said and related services (e.g., along the request path).

TO LOGS

Logs–logs correlation attempts to find related logs by service, environment, or severity. For example, you could ask for all `ERROR` logs from services along the request path.

11.1.5 Correlating profiles

We discussed the nature and use cases of profiles in chapter 9; they are somewhat comparable to log lines, as their scope is a single service. Profiles can be emitted automatically (via eBPF or language-specific means) or manually. In contrast to logs, profiles are numerical. So while many of the preceding to–from correlation observations apply, take into account that profiles are numerical values, at the end of the day (so in a sense, they behave like metrics, from a correlation point of view).